MIL-S-60233C(MU)
1 December 1969
SUPERSEDING
MIL-S-60233B(MU)
28 May 1968

MILITARY SPECIFICATION

SIGHT, REFLEX, HELICOPTER: XM60, XM60E1, AND XM60E2

1. SCOPE

1.1 This specification covers the XM60 Series of reflex sights. Each sight provides a mechanism that may be employed, at the option of the users, to mobilize the sight in increments of elevation and depression. They are fixed in azimuth.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components of Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel: Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

FSC 1270

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DRAWINGS

U.S. Army, Frankford Arsenal

F10541577	Sight, Reflex: XM60
F10553488	Sight, Reflex: XM60E1
F11729760	Sight, Reflex: XM60E2
A11729761	Sight, Reflex: XM60E2 (With Equipment)

PACKAGING DATA SHEETS

10541577	Packaging of Sight, Reflex: XM60
10553488	Packaging of Sight Reflex: XM60E1
11729760	Packaging of Sight, Reflex: XM60E2
11729761	Packaging of Sight, Reflex: XM60E2 (With Equipment)

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.2 Fabrication.- Each sight shall be manufactured in accordance with its applicable drawing specified in Section 2 and drawings pertaining thereto.

3.3 General specifications.- The contractor shall be responsible for adherence to and compliance with the following requirements of MIL-O-13830 and MIL-F-13926.

3.3.1 MIL-O-13830.

(a) Lenses
(b) Prisms and mirrors
(c) Cleanliness

3.3.2 MIL-F-13926.

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

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3.4 Storage at extreme temperatures.- The sight shall not evidence damage after storage at -65° Fahrenheit (F) and +160°F.

3.5 Operability at extreme temperatures.- While thermally stabilized at -40°F and at +150°F, the sight shall meet the mechanical operability requirements of 3.5.1 to 3.5.4.

3.5.1 Electrical components.- Lamp switches and light intensity controls shall demonstrate ease of manual operation between their required limits.

3.5.2 Latch and hinge and filter mechanisms.- The latch and hinge (of the lamp housing assembly) and the filter shall demonstrate ease of manual operation between their required limits.

3.5.3 Click mechanism.- The five mil click mechanism shall disengage with an applied direct force of not more than two pounds greater than the value exhibited when meeting 3.7.2.

3.5.4 Elevation knob torque.- The torque required to rotate the elevation knob shall not be more than five pound-inches greater than that exhibited when meeting 3.7.3.1, throughout the excursions in 3.7.7, with the five mil click mechanism in both an engaged and disengaged state.

3.6 Humidity.- The sight shall not evidence damage after exposure to a 97% relative humidity for 24 hours at +160°F followed by four hours at between +60° and +90°F with condensation taking place on the sight.

3.7 Performance between +60°F and +90°F.- Except as otherwise specified, the sight shall meet requirements of 3.7.2 to 3.7.11 inclusive after orientations "a" to "g" inclusive have been effected (see Figure 1).

a. The plane containing reference line "X" shall be horizontal within 15 mils.

b. The plane containing reference surface "Y" shall be vertical within 5 mils.

c. The circular level vial bubble shall be concentric to the circle of the vial within 1/2 the width of the line forming the circle.

d. The direct current (dc) potential to the sight shall be 28 ± 1.0 volt (v).

e. Each light switch shall be in an "ON" position.

f. Each light intensity control shall be positioned for optimum viewing.

g. The scale of the elevation knob shall indicate zero.

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3.7.1 Vibration (general).- The sight shall evidence no damage or electrical circuit malfunction during or after vibration and throughout vibration exhibit no flickering of the projected reticle image. Subsequent to vibration, the line of sight demonstrated by the sight when meeting 3.7.6 shall not have changed more than one mil.

3.7.1.1 Vibration A.- This vibration shall be for two hours, in each plane of three mutually perpendicular planes, at the sinusoidal frequencies and amplitudes depicted in Figure 2. During each first hour period, the frequency shall be cycled logarithmically from 5 to 500 to 5 cps for 4 cycles each of 15± 1.0 minute duration. A fifteen minute dwell shall occur at each of the most significant resonant frequencies (not exceeding four) exhibited by the sight during the first hour of each two hour vibration in each plane. When meeting this requirement, Scale Lamp MS25237-327 and Reticle Lamp MS16105-2 shall not be subjected to the vibration dwells specified herein.

3.7.1.2 Vibration B.- The sight's projected reticle image shall not be displaced more than two mils throughout this vibration from its initial position prior to vibration. This vibration shall be in a vertical plane for a 10 ± 1.0 minute period at a frequency of 30 cps with a double amplitude of 0.125 inch. This requirement shall be met with Reference Surface "Y" vertical within 5° and Reference line "X" within 5° of a horizontal plane (See Figure 1).

3.7.2 Click mechanism.- Except as otherwise specified in 3.5.3, the direct force required to disengage the five mil click mechanism shall not be less than five pounds nor more than ten pounds.

3.7.3 Elevation knob.

3.7.3.1 Torque.- Except as otherwise specified in 3.5.4, the torque in pound-inches required to rotate the elevation knob through the excursions in 3.7.7 shall not be less than seven nor more than 12 with the click mechanism engaged and not less than two nor more than four with the click mechanism disengaged.

3.7.3.2 Backlash.- Elevation knob backlash shall not exceed one mil throughout the excursions in 3.7.7.

3.7.4 Illumination.

3.7.4.1 Reticle lamp.- In both "ON" positions of the filament selector switch, reticle lamp light intensity shall progressively decrease from maximum to minimum intensity without flicker, when its light intensity control is rotated from an extreme clockwise (cw) to counterclockwise (ccw) position. The reticle lamp shall not illuminate with the filament selector switch in an "OFF" position.

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3.7.4.2 Elevation scale lamp.- In the "ON" position of the elevation scale lamp switch, the elevation scale lamp shall progressively decrease from maximum to minimum intensity without flicker, when its light intensity control knob is rotated from an extreme cw to ccw position. The elevation scale lamp shall not illuminate with its switch in an "OFF" position.

3.7.4.3 Extraneous light emission.- The reticle light source shall not be visible except as observed through the lenses.

3.7.4.4 Projected reticle image definition.- The projected reticle's prime image shall be clearly defined with no visible breaks when viewed from a normal operating position against a bright background (such as the reflection of a clear sky from water) or when viewed in a darkened area. This requirement shall be met with light intensity controls adjusted for optimum viewing of the prime image (see 6.2) and the attenuation of all ghost images.

3.7.4.5 Elevation scale definition.- Elevation scale lamp light intensity shall permit scale numerals and scale index to be clearly distinguishable in a darkened area.

3.7.5 Inclinometer accuracy.- The inclinometer ball shall center within the width of one graduation line.

3.7.6 Collimation.- A line of sight through the center pip of the reticle image on the beamsplitter shall be horizontal within one mil and perpendicular to reference surface "Y" within one mil prior to meeting 3.7.7 to 3.7.11.

3.7.7 Plumb travel.- The line of sight shall track a plumb line within one mil (total deviation) from 180 mils in elevation to 180 mils in depression.

3.7.8 Parallax.- When viewing a target at 1500 ± 500 meters, parallax between the reticle image and target on the optical axis shall not exceed four mils.

3.7.9 Parallelism of reticle and image.- An axis defined by a line passing through the projected horizontal reticle lines shall be parallel to a horizontal target line within 40 minutes of arc.

3.7.10 Projected reticle image accuracy.- The angle subtended by the outer circle of the projected reticle pattern shall be 80 ± 5.0 mils.

3.7.11 Elevation accuracy.- Elevation knob settings between 50 mils elevation and 50 mils depression shall position the line of sight within three mils of the indicated setting. Elevation knob settings between 51 mils and 180 mils elevation, and between 51 mils and 180 mils depression shall position the line of sight within 10 mils of the indicated settings.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-O-13830 and MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) each of the type sight specified in the contract plus three (3) each of all items covered by SQAP which are a part of the specified type. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

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4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAPs, the applicable Quality Assurance Provisions of MIL-O-13830 and MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I and II herein. Examination and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I and II shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENTS	TEST PROCEDURES
CRITICAL: NONE		
MAJOR: AQL 1.0 percent defective		
101. Vibration (general); Vibration B	3.7.1; 3.7.1.2	4.6.4.1; 4.6.4.1.2
102. Click mechanism	3.7.2	4.6.4.2
103. Torque (elevation knob)	3.7.3.1	4.6.4.3
104. Backlash (elevation knob)	3.7.3.2	4.6.4.4
105. Reticle lamp controls	3.7.4.1	4.6.4.5
106. Elevation scale lamp controls	3.7.4.2	4.6.4.6
107. Extraneous light emission	3.7.4.3	4.6.4.7
108. Projected reticle image definition	3.7.4.4	4.6.4.8
109. Elevation scale definition	3.7.4.5	4.6.4.9
110. Inclinometer accuracy	3.7.5	4.6.4.10
111. Collimation	3.7.6	4.6.4.11
112. Plumb travel	3.7.7	4.6.4.12
113. Parallax	3.7.8	4.6.4.13
114. Parallelism of reticle and image	3.7.9	4.6.4.14
115. Projected reticle image accuracy	3.7.10	4.6.4.15
116. Elevation accuracy	3.7.11	4.6.4.16
117. Cleanliness	3.6.1c	4.6.4.17
MINOR: NONE		

NOTE: The tests in Table I shall be conducted in accordance with the test condition requirements in the "Test Facilities" requirement of MIL-F-13926.

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4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Control tests.

4.4.1 Special sampling.- Samples shall be selected from lots at random using inspection level S-3 Table I of MIL-STD-105. Acceptance or rejection of the lot shall be in accordance with AQL 2.5 of Table II-A of MIL-STD-105. The samples shall have met all the requirements and tests outlined in Table I prior to and after being subjected to the following examinations and tests.

TABLE II

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301. Storage at extreme temperature	3.4	4.6.1
302. Operability at extreme temperatures	3.5	4.6.2
303. Electrical components	3.5.1	4.6.2.1
304. Latch and hinge and filter mechanisms	3.5.2	4.6.2.1
305. Click mechanism	3.5.3	4.6.2.1
306. Elevation knob torque	3.5.4	4.6.2.1

4.4.1.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.4.2 First article sample.- Unless otherwise specified in the contract, test(s) listed herein shall be conducted on one unit of the initial production sample and shall not again be performed on the remaining production quantities, provided the manufacturing processes and techniques used to produce the samples have not been changed or revised.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
307. Humidity	3.6	4.6.3
308. Vibration (general); Vibration A	3.7.1; 3.7.1.1	4.6.4.1; 4.6.4.1.1

4.4.3 Certification.- Certification shall include test data and results for all characteristics specified in Table III. Certification shall be provided prior to performing final acceptance inspection, and remain in effect throughout the contract for Government acceptance. This does not relieve the contractor from inspection throughout the contract production, including recording data and results therefrom. Recorded data and results shall be made available to the Government upon request. A new certification will be required should any changes occur in materials, manufacturing processes and techniques previously certified.

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TABLE III

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Materials	3.1	Applicable drawings
Fabrication	3.2	Standards, and
General specification (MIL-F-13926)	3.3.2	Specifications
General specification (MIL-O-13830)	3.3.1 (except 3.3.1c)	

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specification or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents (IEL), when provided with the Technical Data Package (TDP).

4.5.2.2 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.- The test equipment provided in accordance with 4.5.2.2 shall meet the following requirements:

a. Environmental and mechanical test equipment.- Temperature, humidity, and vibration test equipment shall conform to applicable requirements in the "Test Facilities" requirement of MIL-F-13926. Standard vibration equipment provided shall have provisions for controlling the frequency

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and amplitude of vibration in accordance with 3.7.1.1 and 3.7.1.2. Vibration equipment shall have provisions for holding the sight securely throughout vibration tests, and allow its orientation in three mutually perpendicular planes for Vibration A and in one plane only for Vibration B to simulate the sight's position when mounted for use. A standard time measuring instrument shall be provided to control vibration periods. A torque wrench with adapter shall also be provided.

b. Optical test equipment.- Inspection equipment having at least the functional accuracies of Items 1, 2, 3 and 4 in 4.5.2.4 shall permit the sight to be positioned as specified in 3.7. A cross leveling fixture (ref. D6523553) will be utilized in conjunction with an adapter which in turn will support the sight in the position noted above. This adapter shall also position the pivoting axis of the beamsplitter on the same axis as upper shaft axis of the fixture. The adapter will also be provided for holding Collimator (Drawing F5549108) to determine cross level of the fixture. The adapter shall also include a means for supporting a front surface mirror used to autocollimate collimator (Ref. F7573291). The collimator shall contain a suitable reticle (target) containing acceptance or rejection limitation markings and shall be within an accuracy of 0.1 mil. The target vertical line shall be positioned plumb. An adapter to support a collimator (Ref. F7573291) will be incorporated and will allow sufficient adjustments for alignment and autocollimation. A three power (3X) auxiliary telescope shall also be provided.

c. Collimation reticle target.- The target, Figure 3, may employ concentric circles, a vertical line and a horizontal line. Lines shall bisect each other at the center of all circles and line extremities shall exceed the larger circle's diameter. Target lines C and D are provided to measure parallelism between the sight's horizontal reticle lines and target line B. The angular position of the lines shall facilitate compliance with the tolerance specified in 3.7.9. Targets shall contain acceptance limits for optical requirements to be inspected. Target lines A and B intersecting point, when properly positioned in the projector collimator, will represent the optical axis of the collimator. The target, Figure 3, may employ the required series of blocks, each 1 mil square, located about the intersecting points of target lines to facilitate inspection when determining reticle movement, plumb travel, elevation, accuracy and backlash.

d. Electrical test equipment.- A regulated electrical power source (Item 5 in 4.5.2.4) shall be provided to supply the direct current (dc) potential in proper polarity required by individual tests. Electrical instrumentation (Item 6 in 4.5.2.4) shall be provided to measure and monitor the electrical performance of the sight under test, and flexible conductors shall be provided to connect electrical instrumentation installed clear of the vibration table. Electrical instrumentation shall be provided to verify the sight's electrical components and wiring that will not damage circuitry or components.

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4.5.2.4 Special test equipment accuracy.- The special test equipment in 4.5.2.3 shall have at least the accuracy depicted for the following equipment:

ITEM	NOMENCLATURE	DESCRIPTION
1.	Telescope, Collimating for setting Fixtures, Collimating	Depicted on Dwg. F5549108
2.	Cross-leveling Fixture	Depicted on Dwg. D6523553
3.	Collimator, Projector	Depicted on Dwg. F7573291
4.	Inspection Aid (Gunner's Quadrant)	Depicted on Dwg. A8658940
5.	Power Supply 28 V.D.C.	Accuracy of voltage output within 1%, Regulation 8%
6.	Volt-ammeter	0-50 Vdc; 0-1 amp. dc, full scale accuracy ± 2% on both scales

4.6 Test methods and procedures.

4.6.1 Storage at extreme temperatures.- Utilizing test equipment in accordance with 4.5, gradually change chamber temperature to prevent thermal shock until the external surface of the sight is thermally stabilized at each extreme temperature in 3.4. Maintain each extreme temperature for an uninterrupted period of at least 24 hours. Upon completion of each extreme temperature excursion, the temperature shall be gradually returned to between +60°F and +90°F and the sight examined for evidence of damage.

4.6.2 Operability at -40°F and at +150°F.- Utilizing test equipment in accordance with 4.5, gradually change chamber temperature to prevent thermal shock until the external surface of the sight is thermally stabilized at each extreme temperature in 3.5. Maintain extreme temperature for only the uninterrupted period of time required to perform the tests and examinations in 4.6.2.1 at each specified temperature.

4.6.2.1 Tests and examinations at extreme temperatures.- Actuate light switches in all positions to verify ease of manipulation and positive indexing. Manually rotate light intensity controls clockwise (cw) and counterclockwise (ccw) throughout their entire range to verify smooth operation with no binding or slip and that knobs remain secure to their shafts. Fasten, unfasten, and operate the latch and hinge to verify that it exhibits ease of manual operations and secure latching. The filter shall also be manipulated to verify that it exhibits ease of operation and secure positioning. Now engage the five mil click mechanism and affix a standard torque adapter with wrench to the elevation knob. Rotate the elevation knob cw and ccw throughout the range in 3.7.7 to verify that the measured torque is as specified in 3.5.4. Apply to the elevation knob a direct force no greater than that specified in 3.5.3, as measured at point of application, and verify click mechanism disengagement. With the click mechanism disengaged, again measure elevation knob torque to verify that it is not more than the value specified in 3.5.4.

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4.6.3 Humidity.- Except as specified herein, perform this test in accordance with the "Test Facilities" and "Humidity (Procedure III)" requirements of MIL-F-13926. Electrical connector terminals, not exposed when connectors are mated, shall be protected from direct exposure to humidity. Exposure to a relative humidity of 97%, at an ambient temperature of +160°F, for at least 24 hours, shall be followed by a condensation period of at least four hours between +60°F and +90°F - during which time chamber temperature may be adjusted to obtain condensation periods. The humidity phase of this test shall now cease, and the external surface of the test item shall be dried by wiping all accessible areas. Drying by air blast or in an oven shall not be permitted. Immediately after this test, apply a direct current of 28 volts (28 vdc) and verify that no evidence of electrical damage exists prior to meeting the "Post Environmental Inspection" requirement of MIL-F-13926.

4.6.4 Performance between 60°F and 90°F.- The general test requirements and procedures specified in sub-paragraph "a" shall apply when appropriate to meet all requirements of 3.7 that necessitate sight positioning along specific optical or geometric lines.

a. General test procedure.- This general test procedure shall facilitate positioning the sight in compliance with 3.7. Level the cross leveling fixture in all planes with mount adapter assembled to the fixture. Level the fixture by placing Collimator F5549108 in a V slot provided in the adapter (see 4.5.2.3) and by positioning a plumb line approximately 40 feet from the collimator and in coincidence with the viewing telescope's optical axis. Activate the test fixture to raise and lower the line of sight of the telescope. The optical axis shall remain parallel to the plumb line during this activation. Remove the collimator and place the Inspection Aid (Gunner's Quadrant) Drawing A8658940 set to zero over the machine surface provided for this purpose. This surface shall be leveled when the quadrant produces identical level reading when rotated 180° from the original position. Assemble the means provided for autocollimating collimator, Drawing F7573291 to adapter (see 4.5.2.3). Autocollimation of the collimator is performed by sighting through the eyepiece and noting where the reflected reticle line falls with respect to the collimator reticle lines. Proper collimation is assured when the reflected reticle falls upon the collimator reticle lines. Remove the mirror and place the collimator vertical reticle line parallel to a plumb line by rotating the collimator in the supporting medium. Recheck autocollimation. Assemble the unit under test to the adapter (see 4.5.2.3) and observe where the level bubble falls in respect to the level vials circular lines. The bubble must indicate level within the limits specified in 3.7c. Set the elevation knob of the sight to zero elevation by energizing the sight and observing where the center reticle pip falls with respect to the intersecting point of target lines A and B (Figure 3). Record this position.

4.6.4.1 Vibration (general).- Subsequent to compliance with sub-paragraph "a" of 4.6.4, remove the adapter and sight from the test fixture and secure both adapter and sight to a vibration table. Apply 28 vdc to the

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sight's electrical input making the required electrical connections to permit input voltage to be verified and input current monitored. Throughout the entire vibration period, sequentially position the light switches in all positions. Observe no input current change indicative of intermittent electrical circuit continuity, electrical short circuit, or lamp failure when the switches are in an "ON" position, and observe that input current remains at zero when the switches are in an "OFF" position. Examine instrument lamps to determine that they exhibit no flicker in each switch position. Upon completion of this test, the sight shall be given a visual and tactile examination for damage. Reunite the adapter and sight with the test fixture. Energize the electrical system and observe where the center pip falls with respect to the previously recorded setting (4.6.4a) without readjusting either the sight or test fixture. Both readings to be within the limit outlined in 3.7.1. Note where the ball of the inclinometer is with respect to the center position. This position to conform with the requirement specified in 3.7c.

4.6.4.1.1 Vibration A.- Perform this test in compliance with 3.7.1, 3.7.1.1 and 4.6.4.1 and in accordance with Procedure I (Parts 1,2,3) of MIL-STD-810. During this test, set the light intensity controls for maximum illumination of instrument lamps. Throughout the first hour of each two hour vibration in each plane, record all predominant resonant frequencies exhibited by the sight. Upon completion of each first hour of vibration, select the four most predominant resonant frequencies from among those recorded in each plane to acquire a maximum of twelve. During the remaining one hour vibration in each plane, vibrate the sight for at least 15 minutes at each one of the four resonant frequencies previously selected (with lamps removed). This shall result in twelve 15 minute periods - four in each plane - for a total of three hours - in addition to three first hour vibration periods, which altogether shall constitute a six hour vibration period. Lamps shall now be replaced and the sight examined as specified in 4.6.4.1.

4.6.4.1.2 Vibration B.- Position the sight as specified in 3.7.1.2, secure it to the vibration table, and vibrate the sight in a vertical plane only in compliance with 3.7.1.2. Adjust the light intensity controls for optimum viewing, and establish coincidence between the center pip of the projected reticle image and the geometric center of the target reticle. The intersection of the target's cross hair reticle shall be the center of a circle, whose diameter shall equal the chord length of an arc which subtends a four mil angle. Vibrate the sight in compliance with 3.7.1.2 and throughout the entire vibration period, observe no projected reticle image displacement, from the coincidence established herein, in excess of two mils in any direction when the reticle light switch is in either "ON" position. Lamps shall not be removed or replaced during this vibration and shall not fail during this test or during any subsequent examination. The sight shall now be examined as specified in 4.6.4.1.

4.6.4.2 Click mechanism.- With standard measuring equipment, measure the direct force values at the point of application. Increase the applied force until the minimum value in 3.7.2 is attained and verify no prior

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disengagement of the five mil click mechanism. Also verify that the direct force required to obtain disengagement is not in excess of the maximum value in 3.7.2.

4.6.4.3 Elevation knob torque.- Use a standard torque wrench and adapter for this test. Affix the adapter and wrench to the elevation knob, and rotate the elevation knob in a clockwise and counterclockwise direction throughout the ranges in 3.7.7 with the five mil click mechanism in an engaged mode and then in a disengaged mode. At least three readings shall be taken in each mode to verify compliance within the torque limits specified in 3.7.3.1 for each mode.

4.6.4.4 Elevation knob backlash.- This test shall be performed after determining accuracy of elevation, in accordance with 4.6.4.16. The test is performed by approaching the previously selected setting set into the sight and cross leveling fixture from the opposite direction. Exercise care not to override the point of coincidence between the reticle center pin and selected target point. Backlash is to be considered as the difference in setting as read on the inspection aid when approaching the recorded setting from the opposite direction. This difference in reading shall not exceed the limits outlined in 3.7.3.2.

4.6.4.5 Reticle lamp controls.- Apply 28 vdc to the sight's electrical input and monitor input current throughout this test. Place the reticle light switch in an "ON" position, rotate the reticle light intensity control knob cw from stop to stop, and observe a progressive increase in the intensity of reticle lamp emitted light. Place light switch in the alternate "ON" position, rotate light intensity control knob ccw from stop to stop, and observe that the intensity of emitted light will progressively attenuate to minimum. Emitted light and input current shall be under constant scrutiny during this test for evidence of circuit or component malfunction. Place light switch in the "OFF" position and observe that input current remains at zero regardless of light intensity control knob position. Determine that light intensity control knob is secure to its shaft and that the light switch exhibits positive indexing in all three positions.

4.6.4.6 Elevation scale lamp controls.- Apply 28 vdc to the sight's electrical input and monitor input current throughout this test. Place scale light switch in an "ON" position, rotate scale light intensity control knob cw and ccw from stop to stop, and observe that the intensity of emitted light will progressively increase with cw rotation and decrease with ccw rotation. Emitted light and input current shall be under constant scrutiny during this test for evidence of circuit or component malfunction. Place light switch in the "OFF" position and observe that input current remains at zero regardless of light intensity control knob position. Determine that light intensity control knob is secure to its shaft and that the light switch exhibits positive indexing in both positions.

4.6.4.7 Extraneous light emission.- With the sight in a darkened room (illumination not to exceed 0.01 foot candle), adjust the reticle light intensity control knob to obtain maximum filament intensity. No portion of

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the non-opaque surface of the lamp shall be visible from any point except through the lenses. This test shall be conducted with the elevation scale lamp switch in the "OFF" position.

4.6.4.8 Projected reticle image definition.- Rotate the light intensity control knob to obtain optimum resolution of the projected reticle pattern image in a darkened room (as specified in 4.6.4.7). The observer shall view the projected reticle's prime pattern image through a 3X auxiliary telescope to determine that it is clearly defined with no breaks. After a lapse of at least 15 minutes - a light adaptation period - the light intensity control shall again be adjusted to obtain optimum resolution of the projected reticle's prime pattern image against a uniformly illuminated field of view. The field of view shall have a brightness of not less than 2300 foot lamberts. Verify that a position exists where the projected reticle pattern image is clearly visible.

4.6.4.9 Elevation scale definition.- Place the elevation scale light switch in an "ON" position, and in a darkened room (as specified in 4.6.4.7), adjust the elevation scale light intensity control knob for optimum resolution of elevation scale numerals and markings. The examiner shall verify that all elevation scale numerals and markings are readable from a normal operating position when viewed with the unaided eye (corrected as required to provide a capability not in excess to that demonstrated by 20-20 vision).

4.6.4.10 Inclinometer accuracy.- Utilize the general test procedure in subparagraph "a" of 4.6.4 and observe compliance with 3.7.5.

4.6.4.11 Collimation.- Prior to conducting tests of 4.6.4.12 and 4.6.4.16, utilize the general test procedure specified in subparagraph "a" of 4.6.4 and observe compliance with 3.7.6.

4.6.4.12 Plumb travel.- Utilize the general test procedure specified in subparagraph "a" of 4.6.4 and the position recorded prior to vibration. Rotate the elevation knob in one direction and the elevation hand wheel of the cross leveling fixture an equal amount in the opposite direction. Prior to and during this rotation, sight through the beamsplitter, and observe coincidence between the center pip and a selected mil block on target (Figure 3). This coincidence shall not exceed the limit specified in 3.7.7 when the center pip is moved 180 mils in elevation and 180 mils in depression from the position recorded prior to vibration.

4.6.4.13 Parallax.- Utilize the general test procedure in subparagraph "a" of 4.6.4, and while sighting through the beamsplitter, observe coincidence between a selected target point (previously established in "a" of 4.6.4) and the center reticle pip. The observer shall now move his head from right to left while observing that the coincidence stated herein is retained, within the limit specified in 3.7.8, when viewing a target at an actual or simulated distance of 1500 meters.

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4.6.4.14 Parallelism of reticle and image.- Utilize the general test procedure in subparagraph "a" of 4.6.4 and an auxiliary telescope having a field of view (FOV) greater than 85 mils (whose horizontal reticle line is level). Establish coincidence between the center pip of the sight's reticle image and the intersection of the cross hair pattern on both the auxiliary telescope's reticle and on a target whose horizontal line has been previously made level. Over an 85 mil FOV, examine each extremity of the sight's horizontal reticle line image and verify the existence of no displacement greater than 0.21 milliradian. Lines "C" and "D" on the target depicted in Figure 3 will facilitate this examination.

4.6.4.15 Projected reticle image accuracy.- Rotate the light intensity control knob to obtain optimum resolution of the projected reticle pattern image. Utilize two target circles having the required diameters. Circle diameter shall equal the chord length of an arc which subtends a 75 mil and 85 mil angle. Establish coincidence between the geometric center of the target circle and the center pip of the projected reticle pattern image. Observe in all four quadrants the periphery of the image is within the tolerance specified in 3.7.10.

4.6.4.16 Elevation accuracy.- Utilize the general test procedure specified in subparagraph "a" of 4.6.4 and the position recorded prior to vibration. From this position set the elevation knob 25 mils in depression. With the inspection aid set to 25 mils, position it on the appropriate surface of the fixture (to indicate following setting of the fixture). Adjust the cross leveling fixture in elevation to center the level vial bubble on the inspection aid. Observe the position of the line of sight in elevation. The difference between this reading and originally recorded reading shall be within the limits specified in 3.7.11. Repeat this procedure with the elevation knob set for 50, 51, and 180 mils depression; 25, 50, 51, and 180 mils elevation; with the inspection aid appropriately set and positioned for these tests. This test should be conducted subsequent to removal of elevation movement backlash.

4.6.4.17 Cleanliness.- Subsequent to conducting all tests and examinations, the optical surfaces of the sight shall be examined in accordance with MIL-O-13830 to determine compliance with 3.3.1(c).

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with the applicable Packaging Data Sheet specified in Section 2. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- Reflex sights covered by this specification are primarily intended for use as sighting devices on helicopters. They are intended to increase the effectiveness of weaponry (2.75 rockets) and be

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part of the XM16, XM21, and XM156 Armament Subsystems. When energized by the aircraft's electrical system, the sights are intended to superimpose a projected reticle pattern image on a target to provide a reference for estimating lead angles.

6.2 Ghost images.- Multiple reflections are unavoidably generated between the two surfaces of the beamsplitter and these reflections form ghost (secondary) images of the reticle. These images cannot be eliminated but their intensity can be minimized by adjusting the reticle lamp light intensity control. The projected reticle image specified as the prime image in 3.7.4.4 will be the brightest image in reference to all other visible images and it will be utilized when meeting all requirements of this specification wherein image observation is required.

6.3 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification
b. Requirement for First Article Testing
c. Applicable packaging data sheet number (see 5.1)
d. Selection of applicable levels of preservation, packaging and packing
e. Applicable stock number and type of sight to be procured

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1270-A036

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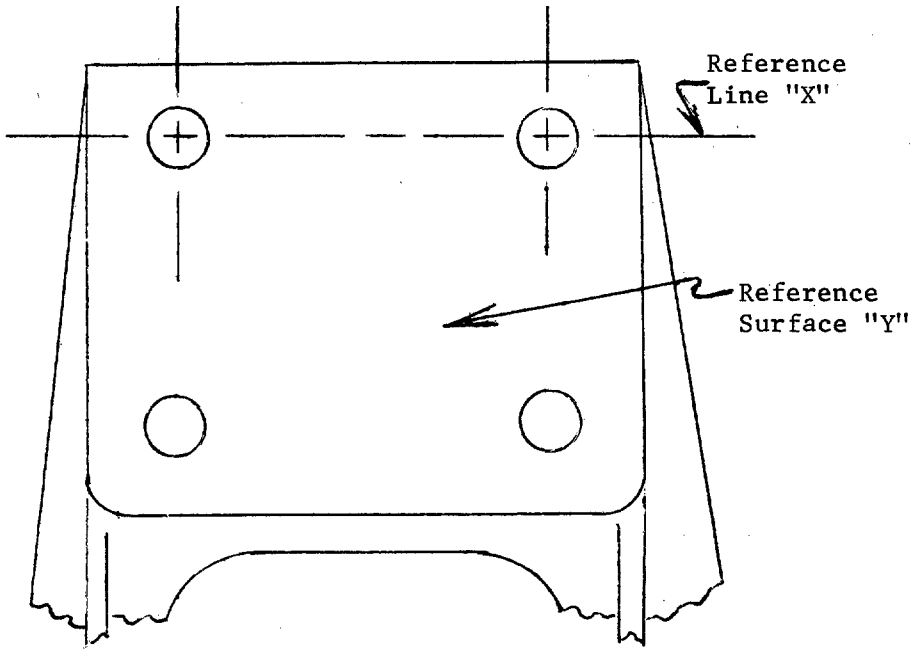


FIGURE 1.

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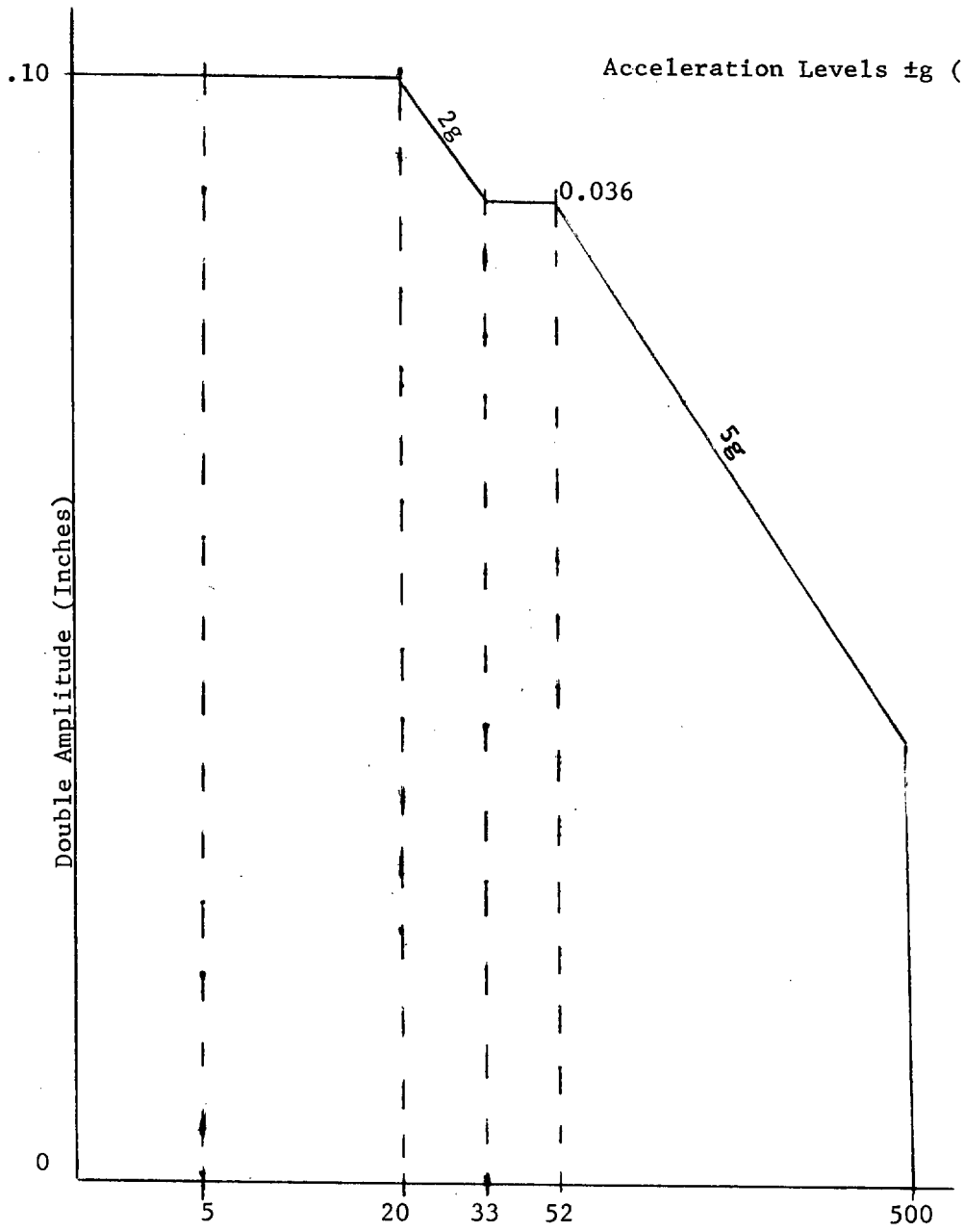


FIGURE 2. Vibration "A" Curve

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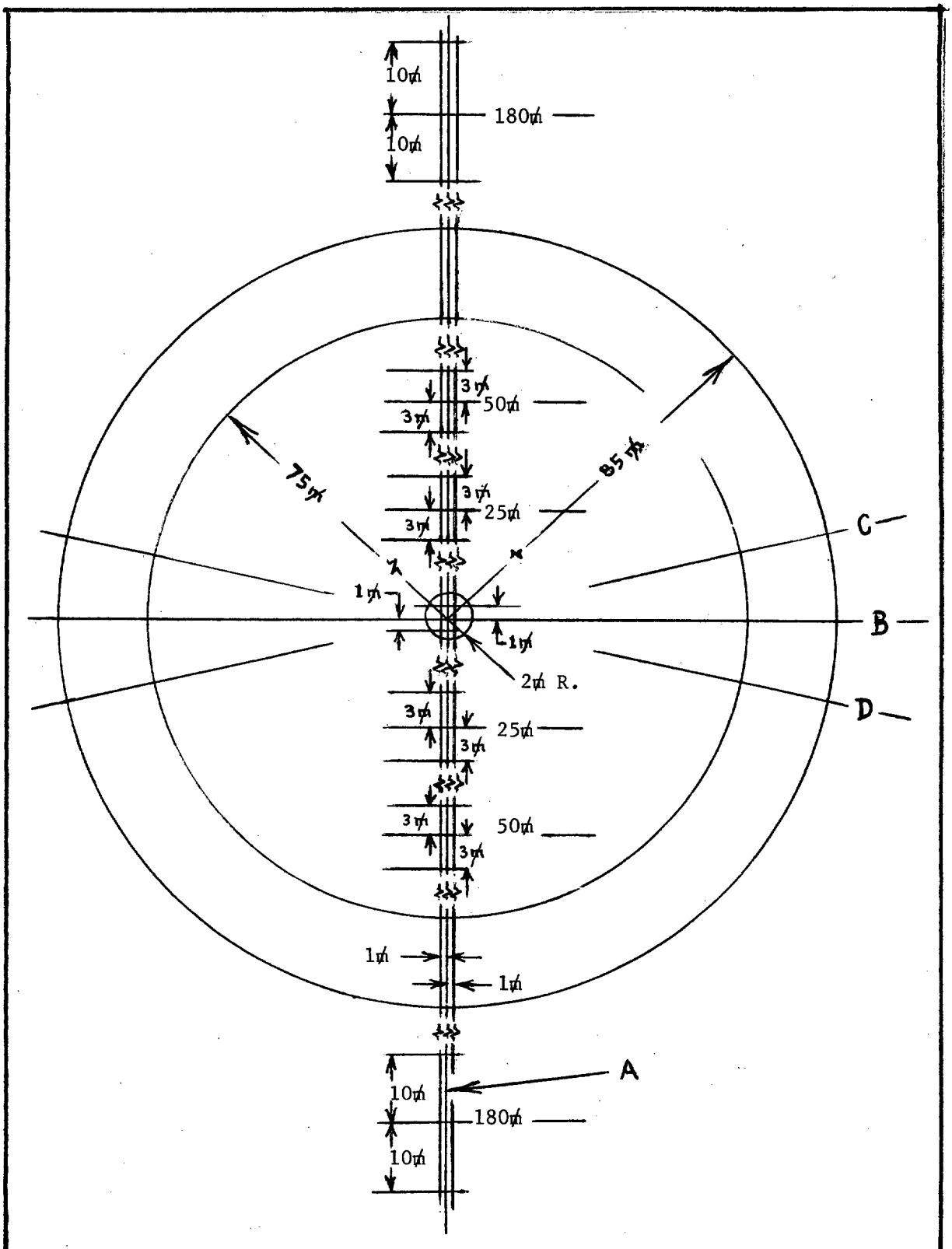


FIGURE 3.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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